

JavaScript Based Path Tracer Simulator

SRIDHAR BUKYA

Path tracing is commonly seen in robotics, where a robot is programmed to follow a path automatically without any external commands. Such robots are generally termed as line-following robots. Some of the robots also ply along a pre-

versed by the user using a mouse cursor or any pointing device connected to the computer console.

Software program

The software program (PTS.htm) is coded in JavaScript using Notepad. When the program runs, Internet Explorer shows up a screen as shown in

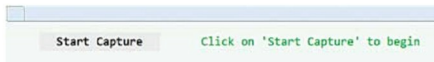


Fig. 1: Program output screen



Fig. 2: Red-colour chili as the object

traced path with precise accuracy.

This program does not go deep into robotics but simulates the line-following robot principle by following the path first tra-

ced. The simulator has an all-in-one button that is programmed to accomplish three tasks: starting, stopping and saving the mouse path capture.

The user has to first click 'Start Capture' button to start capturing the movement of mouse cursor. Within a second this button changes dynamically to function as 'Stop Capture' button. During this time the user has to move the mouse on the screen as desired.

When the user clicks 'Stop Capture' button, the button changes its function to 'Save.' On clicking it, the program gets automatically saved on the hard disk in the path as already defined in the code. When you open the saved program, you will see an animated object following the path of the mouse cursor on the screen. Here a red-colour chili is used as the object (refer Fig. 2).

Main functions of the program

Array(). The pre-defined Array() function is used to store coordinates of the path traced by the movement of

the mouse.

Animate(). It initialises the elements before capturing takes place.

Jump(). It tracks the values of 'x' and 'y' coordinates of the mouse as it moves along the screen. This function is called by the Array() function every time and changes the 'red chili' path dynamically.

Save_RAM(). This function temporarily saves the path traced.

Script_Code(). It dynamically creates file contents that get a call from the Save_HDD() function for permanently saving the file to system hard disk.

Start(). This function dynamically changes the button property and its functionality among three preset functions.

Testing

The program was tested on a Sony VAIO notebook running Windows 7 Home Basic Edition (64-bit) with 3GB DDR3 RAM, 320GB HDD, Internet Explorer 8 and JavaScript plug-in enabled. The program was tested successfully without any runtime error.

Before running the program, make sure that all the ActiveX control and Internet security options are enabled as shown in Fig. 3. Now proceed as follows:

1. Open PTS.htm with Internet Explorer.
2. Click 'Start Capture' button to capture mouse cursor movement on the screen. Move the mouse cursor from top to bottom of the screen.
3. Click 'Stop Capture' button to stop the action. A warning message will appear on the screen as shown

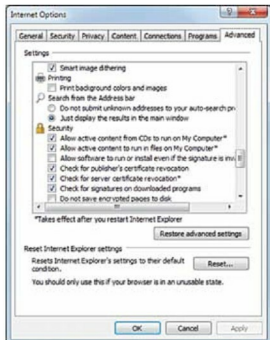


Fig. 3: Internet security options

